

CUSTOMER BEHAVIOR ANALYSIS USING MACHINE LEARNING

1. Project Objective

The objective of this project is to analyze customer transaction and behavior data to identify customer segments, purchase patterns, and churn risks. The project uses data cleaning, feature engineering, exploratory data analysis (EDA), and K-Means clustering to group customers with similar purchasing behavior. The insights obtained from this analysis help businesses improve customer engagement, reduce churn, and make data-driven marketing decisions.

2. DATASET INFORMATION

Dataset Name	Customer Behavior Analysis
Source	Kaggle
Number of Records	250,000
Number of Features	13
Target Column	Churn

3. DATA CLEANING

The dataset was loaded into Google Colab using Pandas. Before analysis, the dataset was cleaned to improve data quality and ensure accurate results.

- Checked the dataset structure using `df.info()`
 - Verified the dataset dimensions and column names.
 - Checked for missing values using `df.isnull().sum()`.
 - Removed duplicate records.
 - Converted the Purchase Date column to datetime format.
 - Filled missing values in the Returns column with 0.

df.info

Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	Customer ID	250000 non-null	int64
1	Purchase Date	250000 non-null	object
2	Product Category	250000 non-null	object
3	Product Price	250000 non-null	int64
4	Quantity	250000 non-null	int64
5	Total Purchase Amount	250000 non-null	int64
6	Payment Method	250000 non-null	object
7	Customer Age	250000 non-null	int64
8	Returns	202404 non-null	float64
9	Customer Name	250000 non-null	object
10	Age	250000 non-null	int64
11	Gender	250000 non-null	object
12	Churn	250000 non-null	int64

dtypes: float64(1), int64(7), object(5)
memory usage: 24.8+ MB

df.isnull().sum()

Customer ID	0
Purchase Date	0
Product Category	0
Product Price	0
Quantity	0
Total Purchase Amount	0
Payment Method	0
Customer Age	0
Returns	47596
Customer Name	0
Age	0
Gender	0
Churn	0

4.FEATURE ENGINEERING

- **Purchase Month** – Extracted the month name from the Purchase Date.
- **Purchase Year** – Extracted the year from the Purchase Date.
- **Age Group** – Categorized customers into different age groups.
- **Purchase Level** – Classified customers as Low, Medium, High, or Premium based on Total Purchase Amount.
- **Churn Status** – Converted churn values into Active and Churned customer labels.

```
df["Purchase Date"] = pd.to_datetime(df["Purchase Date"])
```

```
df["Purchase Month"] = df["Purchase Date"].dt.month_name()
```

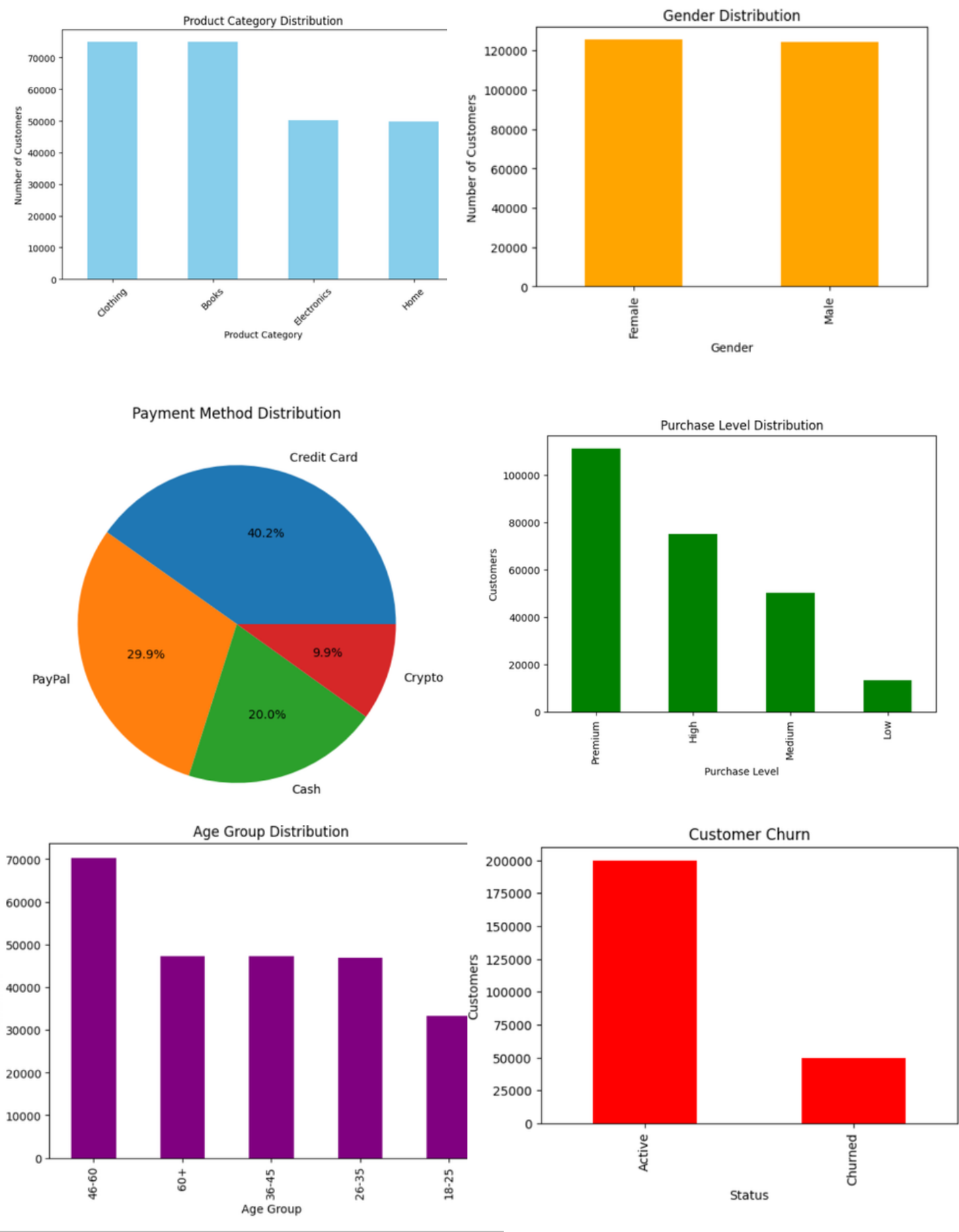
```
df["Purchase Year"] = df["Purchase Date"].dt.year
```

```
df["Age Group"] = pd.cut(  
    df["Age"],  
    bins=[18,25,35,45,60,100],  
    labels=["18-25","26-35","36-45","46-60","60+"]  
)
```

```
df["Purchase Level"] = pd.cut(  
    df["Total Purchase Amount"],  
    bins=[0,500,1500,3000,df["Total Purchase Amount"].max()],  
    labels=["Low","Medium","High","Premium"],  
    include_lowest=True  
)
```

```
df["Churn Status"] = df["Churn"].map({  
    0: "Active",  
    1: "Churned"  
})
```

5. Exploratory Data Analysis(EDA)

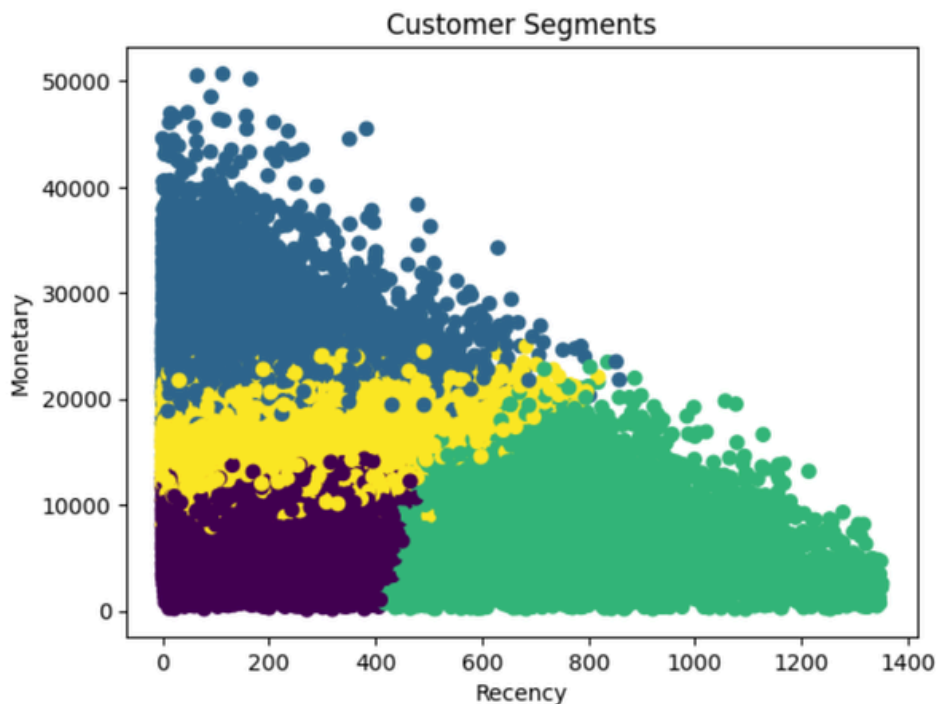


6. Customer Segmentation (K-Means)

Customer segmentation was performed using the K-Means clustering algorithm. The model grouped customers into four clusters based on their purchasing behavior using the following features:

- Age
- Product Price
- Quantity
- Total Purchase Amount

The numerical features were standardized using StandardScaler before applying K-Means to ensure equal importance for each feature.

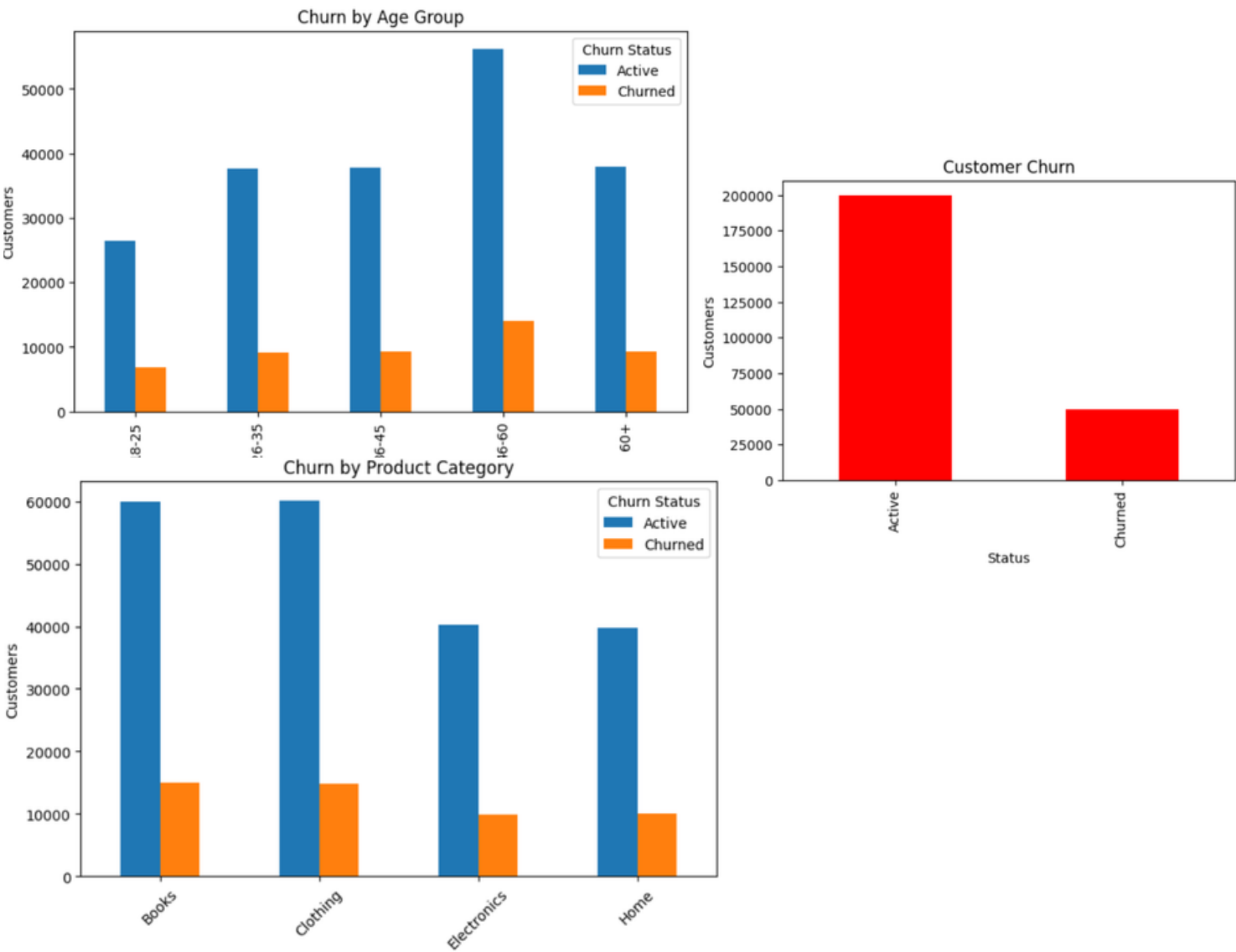


Key Observation

- Customers were successfully grouped into four distinct segments.
- High-value customers contribute significantly to total revenue.
- Medium-value customers have the potential to become premium customers.
- Customer segmentation helps in designing personalized marketing strategies.

7. CHURN ANALYSIS

Customer churn analysis helps identify customers who are likely to stop purchasing. Understanding churn enables businesses to improve customer retention through targeted strategies.



Key Observation

- Most customers remain Active, while a smaller proportion are Churned.
- Customers in the 46–60 age group show relatively higher churn.
- Some product categories experience more churn than others.
- These insights can help businesses design targeted retention strategies.

8. KEY FINDINGS

- The dataset contains 250,000 customer transaction records.
- Data cleaning improved the quality of the dataset.
- Feature engineering created meaningful variables for analysis.
- Customer segmentation identified four distinct customer groups.
- Churn analysis highlighted customer retention opportunities.

9. RECOMMENDATIONS

- Implement loyalty programs to increase repeat purchases.
- Provide personalized product recommendations.
- Offer special discounts to customers at risk of churn.
- Target marketing campaigns to high-value customer segments.
- Regularly monitor customer behavior and update retention strategies.

10. CONCLUSION

This project successfully analyzed customer behavior using data cleaning, feature engineering, exploratory data analysis, and K-Means clustering. The analysis identified customer segments, purchasing patterns, and churn trends. The insights and recommendations can help improve customer engagement, reduce churn, and support better business decisions.

11. TOOLS & LIBRARIES USED

